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Business simulation games integrated with emerging technologies in business education: A review

Aliyev Murad[a,*], Fatullayev Vagif[b], Aliyeva Sadagat[c]

[a] School of Educational Technology, Faculty of Education, Beijing Normal University, Room 118, Yanbo Hall, Beijing 100875, China
[b] Department of Economics and Management of Education, Institute of Education of the Republic of Azerbaijan, 86, Afiyaddin Jalilov street, Baku AZ1001, Azerbaijan
[c] Faculty of Economics and Management, Sumgait State University, Baku Street, 1, Sumqayit AZ5008, Azerbaijan

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ABSTRACT

Business simulation games (BSGs) have gained popularity in higher education, offering students hands-on experiences. Recent advancements in emerging technologies have further fuelled their growth, contributing to the global expansion of BSG market. This review aims to examine the integration of AI, generative AI chatbots, extended reality (XR), virtual reality (VR) (semi-immersive – 3D animated; fully immersive – business training simulators; mobile VR – 360-degree videos), blockchain, mobile and web-based technologies into BSGs and their potential to enhance learning outcomes. This literature review was conducted following the PRISMA framework. Through thematic synthesis analysis of 46 empirical studies, this study identified three major themes: (1) the pedagogical benefits of integrating emerging technologies in BSGs; (2) the factors that shape active student participation, including contextual relevance, instructional design, and intrinsic motivation; and (3) the persistent challenges related to technology integration, cultural adaptability, and inclusive access. The review highlights the dominance of quantitative and mixed-methods approaches, a lack of research in non-Western contexts, and the underrepresentation of long-term impact studies. It also emphasizes the need for pedagogically grounded, culturally responsive, and inclusive BSG designs supported by well-prepared instructors.

1. Introduction

Business simulation games (BSGs), which provide practical and experiential learning opportunities to develop students' decision-making, strategic thinking, and problem-solving skills in a realistic yet risk-free environment, are of great importance in the learning process [57,61,64,66]. The growing demand for these games is confirmed by recent statistics that the market of BSGs is expected to reach $5.8 billion by 2032 at a compound annual growth rate (CAGR) of 9.5 percent [56]. The rapid advancement of business simulation tools is driven by technological innovations that enhance realism and interactivity. Artificial intelligence (AI) enables adaptive, data-driven scenarios that respond to user decisions, providing students with a dynamic and practical learning experience [19]. Immersive technologies such as virtual reality (VR), augmented reality (AR), and extended reality (XR) create interactive environments that replicate real-world challenges, fostering deeper engagement [24,47,48,61,66]. Additionally, mobile and web-based platforms have improved accessibility, allowing learners to participate in simulations anytime and anywhere. These developments are transforming business education by bridging theoretical knowledge with hands-on application, equipping students with the critical skills required in today's competitive job market [11,41,66]; (Jiang et al., 2023).

While studies have explored emerging technology-integrated BSGs from different aspects such as user interface design, learning outcomes, and technological affordances [16,18,19,22,48,71], there remains a notable gap in evaluating how these technologies are effectively integrated and adapted across diverse educational and cultural contexts. Thereby, an up-to-date review is crucial to examine their potential benefits or challenges as well as influencing factors in active learning and student engagement. Moreover, tackling issues such as accessibility, scalability, and the demand for personalized learning experiences is crucial for promoting the wider adoption of these innovative games [10,12,21,23,29,53,64,73]. To address this, the present study seeks to bridge this gap by synthesizing existing literature on BSGs that incorporate emerging technologies, and by exploring the following research questions:

* Corresponding author.
E-mail addresses: aliyev.murad@mail.bnu.edu.cn (M. Aliyev), vaqif.fatullayev@arti.edu.az (V. Fatullayev), sadaqat.aliyeva@sdu.edu.az (S. Aliyeva).

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1. What are the methodological trends and research contexts of the reviewed studies?
2. What are the benefits of incorporating emerging technologies with business simulation games for students' active learning?
3. What are the limitations of business simulation games incorporated with emerging technologies?
4. What factors encourage students to actively participate in business simulation games integrated with emerging technologies?

2. Literature review

2.1. Business simulation games

Business simulation games (BSGs) are interactive, experiential learning tools that replicate real-world business environments, allowing participants to make decisions, solve problems, and manage resources in a risk-free setting. According to Faria et al. (2009), BSGs are "computer-based models of business operations that enable learners to experiment with different strategies and observe the outcomes of their decisions in a simulated environment." These games are designed to enhance critical skills such as strategic thinking, decision-making, and teamwork, making them a valuable pedagogical tool in business and management education [57,61]. BSGs often incorporate elements of gamification, such as scoring systems, leaderboards, and feedback mechanisms, to increase engagement and motivation among learners [66].

Over the decades, advancements in technology such as artificial intelligence, machine learning, and cloud-based platforms have enabled the integration of real-time data, complex algorithms, and interactive interfaces, making BSGs increasingly relevant for modern business training and education. BSGs have expanded to encompass a wide range of business-related disciplines, including finance, accounting, marketing, operations management, human resources management, entrepreneurship education [26,64]. Notable examples include *The Business Strategy Game* and *Capstone,* which focus on market dynamics and competitive positioning, and *The Beer Game*, which illustrates supply chain complexities [18,52]. They enhance students' understanding of economic principles and market dynamics. As a consequence, students engage in complex decision-making scenarios that mirror real-world business challenges.

3. Emerging technologies and business simulation games

The rapid development of innovative technologies makes it challenging to define which innovations qualify as emerging technologies. Because some innovative technologies are widely utilized in developed regions but not yet adopted in many others. Nevertheless, experts attempt to classify certain technologies as emerging, focusing on those that are widely used yet still evolving [1,5]. In this regard, Horizon Report's categorization is crucial for adopting these technologies in higher education institutions across globe. According to this report, emerging technologies are as following: Artificial intelligence (AI) /machine learning (ML), learning analytics (LA), immersive learning technologies (e.g., VR, AR, XR, MR), robotics, 3D modelling software and Open Educational Resources (e.g., simulations, web-based or mobile-based platforms, etc.) [1,5,42].

The researcher Küfeoğlu [42] broadened this horizon providing more technology types such as blockchain, big data, internet of things (IoTs), natural language processing (NLP), cloud computing. Additionally, Küfeoğlu's [42] categorize virtual reality (VR) technologies in three groups: non-immersive, sensory-immersive, and neural-direct. In non-immersive VR, users interact with virtual environments through a standard desktop setup, like pilots using flight simulators for training. Sensory-immersive VR involves larger setups like the CAVE system, where users experience projected 3D visuals, such as architects navigating a building model. Finally, neural-direct VR connects the brain directly to a virtual environment, offering a fully immersive experience, such as future brain-computer interfaces in medical training. All in all, the use of these innovative technologies in education aims to improve accessibility, foster engagement among learners, and provide personalized learning tailored to the needs and learning styles of students.

3.1. Theoretical framework

Theoretical foundations of BSGs are a powerful tool for preparing students for real business processes in educational settings. Several theoretical frameworks exist to explain their effectiveness. The three main pillars—Kolb's experiential learning model (1984), Vygotsky's constructivist approach (1978), and Deci-Ryan's self-determination theory (1985)—serve as a comprehensive guide to how BSGs should be designed and implemented in the classroom.

Experiential learning theory from Kolb [37] defines learning process in four stages – concrete experience → reflective observation → abstract conceptualization → active experimentation—are fully realized in BSG scenarios. For example, in a simulation called Marketplace Live, students manage a startup entering the global smartphone market, making their own pricing, marketing, and production decisions [69]. In the next stage, they look at sales figures and discuss the reasons (reflection), then model strategies to increase market share (abstraction), and finally implement these strategies in the new quarter (testing). Research shows that such a cycle strengthens cognitive structure — especially the mental models that shape decision-making [8,17,55].

According to Vygotsky [75], knowledge is "constructed" in the process of active action and social dialogue. BSGs follow this principle and encourage students to solve problems in teams dynamically. For example, in The Beer Game simulation [36], students who act out different "links" of a logistics chain interpret an order signal and collectively evaluate delivery delays and the impact of different decisions on the system. Social interaction allows students to defend their arguments and learn from the perspective of their peers, which is a key condition for constructivist learning [58].

According to Deci and Ryan's [14] model, feelings of autonomy, competence, and connectedness enhance learners' intrinsic motivation. In BSGs, students are more motivated because they can make independent decisions (autonomy), see results in real time (competence), and are in constant communication with their team members (connectedness). In the CAPSIM Core simulation, for example, student teams monitor financial performance live on a dashboard, exchange ideas with rival teams in a forum, and freely choose strategies for the next round [17,26]. These design features increase student motivation and sustain learning outcomes in the long term.

Thus, the Kolb cycle provides students with the opportunity for systematic reflection, the constructivist environment supports deep learning through social discussion, and self-determination provides ongoing motivation. The integration of all three theoretical approaches makes BSGs a powerful pedagogical tool that allows abstract business concepts to "live" in a real-world context, as well as developing students' analytical, communicative, and strategy-building skills.

3.2. Previous review studies on business simulation games

The reviews conducted provide findings that reflect the realities of business simulation games and their effectiveness. These reviews have examined simulation games from the perspective of pedagogical, technological, methodological, and evaluation practices. This section, in addition to presenting the conclusions of these studies, has highlighted the shortcomings of these articles. By summarizing these studies, as well as providing a comprehensive understanding of the advances and challenges in BSGs, it has identified certain gaps for ongoing research. The following is the analysis of the previous review studies:

Although Ke [35] provided a theoretical analysis of BSGs, the study did not provide examples from case studies and did not address the shortcomings of the game design. Additionally, the analysis did not discuss the assessment practices during simulation games comprehensively. In contrast, Ferreira et al.'s [18] analysis focused on simulation games designed with human–computer interfaces to identify the effects of real-time feedback in participants' engagement. They identified real-time feedback in the games as an important factor. However, this study lacked the discussion on technological constraints and the alignment of simulation games with learning objectives. Likewise, a review conducted by Vlachopoulos and Makri [72] found that simulation games provide students' active participation in the learning process and strengthen their knowledge and skills. Researchers emphasized the need to align the simulation games with learning objectives and course curriculum. Nevertheless, this study has not yet addressed how BSGs can be adapted to different student groups and learning styles.

Scholtz and Hughes [62] reviewed pedagogical interventions in the simulation-based learning. These interventions included didactic interventions, preparatory activities, reflective practices, coaching, mentoring, team structuring, assessments, and promoting collaborative learning and engagement. Given that this review covers articles published between 2007 and 2017, it is inevitable to review more recent research. Furthermore, the rapid discovery of emerging technologies in recent years suggests that updated reviews may directly address emerging issues in BSGs. Similar to this review, Hallinger and Wang [25] extended the timeline further, tracing the development of simulation-based learning in various fields from 1965 to 2018. This study found that fields such as medicine and business increasingly employed different simulation games. Nevertheless, the researchers suggested to conduct more research to examine the effectiveness of pedagogical methods.

Faisal et al.'s [17] systematic review analysed the importance of BSGs in university students' knowledge and skill development. This review showed that the games improve critical thinking, teamwork, and decision-making skills in those who play them. However, the study did not analyze standardized assessment methods to assess the results of the games. On the other hand, although the short-term benefits of BSGs are well-established, their long-term impact on career development and professional skills remains unexplored. As a result, it is crucial to investigate the contributions of these games to students' ongoing professional development and behavior. In addition to this review study, Vélez et al. [71] also explored the impact of simulation games on intrinsic motivation. Researchers found that these games could boost intrinsic motivation and thus encourage participants to actively participate in the games. They found that these games, when closer to a real business environment, develop competition and interaction in participants. However, the study did not consider the effects of simulation games on different learning outcomes and decision-making skills. Following these review findings, Fu et al.'s [20] study also explored that BSGs, particularly those played on personal computers, had a positive impact on learning outcomes and engagement among adults aged 18 and older. The study used mainly survey and quasi-experimental methods. However, the review did not include longitudinal studies, so the long-term effects of business games were not fully explored. At the same time, the review did not fully address practical issues such as costs and integration into the curriculum, as well as technological developments (e.g., VR/AR, artificial intelligence) and pedagogical approaches such as developing students' soft skills through games.

Regarding the scope of the BSG-related articles, Bach et al. [6] divided them into three groups: learning-driven, domain-driven, and technology-driven articles. The study showed that the most cited articles are mainly learning-driven, focusing on teaching, learning and knowledge transfer, and domain-driven, which touch on topics such as decision-making and entrepreneurship. In contrast, the analysis showed that few studies were existed from a technology-driven perspective, particularly demonstrating advances in artificial intelligence (AI), augmented reality, and e-learning technologies.

Tchokoté and Bawack [67] conducted a literature review and bibliometric analysis to examine the relationship between business games and business and entrepreneurship education. The study identified the structure of knowledge and emerging trends in the field, highlighting the importance of college students, learning environments, and experiential learning. As well as, the review observed a significant increase in students' entrepreneurial intentions during engaging in simulation games. At the same time, the research in this area appeared fragmented, pointing to the need for more focused studies and better integration of existing findings.

Review studies related to BSGs show that these educational games improve learning outcomes, provide motivation and active participation of students, and facilitate the integration of technological tools into education. Although existing reviews have evaluated these simulation games from multiple aspects, the integration of the advanced technologies into BSGs necessitates the evaluation of the emerging simulation games in the learning process. Regarding the previous systematic review studies, although the potential of artificial intelligence and immersive learning technologies (e.g., AR, VR) is recognized, their practical application in educational settings is understudied. It is necessary to conduct more up-to-date reviews, as some new interventions may not be included in the existing literature. Likewise, a major shortcoming is the lack of consistent and standardized tools to evaluate the effectiveness of BSGs that incorporate advanced technologies.

4. Methodology

4.1. Literature search strategy

This study conducted a literature review adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework [54]. To ensure a comprehensive and reproducible search strategy, Boolean operators were employed, incorporating a structured combination of keywords (e.g., AND, OR) to capture multiple dimensions of the topic. To ensure full transparency, we reproduce the composite Boolean expression that underpinned all database queries; the verbatim strings for each platform are provided in Appendix A. The search process was systematically designed to identify relevant literature covering three key thematic areas: (1) business simulation games, (2) emerging technologies, and (3) business and management education. To achieve this, keywords and their synonymous terms were carefully selected, tested, and refined based on an iterative approach that considered variations in terminology across different research domains. The final set of search parameters included:

- Business simulation games: "Business simulation games" OR "BSG" OR "management simulation games" OR "serious games" OR "business simulations"
- Management education: "Business schools" OR "management education" OR "higher education institutions" OR "MBA programs" OR "Accounting" OR "Finance" OR "Marketing"

Moreover, when it comes to emerging technologies, these technologies are already widely adopted in developed regions but remain underutilized in other parts of the world. Nevertheless, experts classify certain technologies as emerging based on their ongoing evolution and increasing adoption in education and training contexts [1,5,42]. The emerging technologies search included keywords to capture Internet of Things and blockchain studies. For example, we included *"Internet of Things"* or *"IoT"* and *"blockchain"* to retrieve research on connected devices or secure credentialing systems. Other technology-related terms covered artificial intelligence (e.g. *"AI-driven simulations"*), learning analytics, virtual/augmented/extended reality (e.g. *"VR"*, *"AR"*, *"XR"*), cloud computing, NLP, robotics, automation, 3D modeling software, open educational resources (OER) and digital platforms – web- and mobile-based platforms. This iterative keyword strategy was refined through pilot searches to balance breadth and specificity. Although IoT and blockchain were less represented in the final pool of included studies, they were deliberately embedded in the initial search strategy to ensure that emerging and less-explored technologies were not overlooked. Their inclusion reflects the study's commitment to comprehensive coverage and future-oriented educational research.

The search was applied to five academic databases: Web of Science, Scopus, ERIC (Education Resources Information Center), IEEE (Institute of Electrical and Electronics Engineers), and ACM (Association for Computing Machinery) Digital Library. We conducted initial exploratory searches and adjusted terms with low relevance (e.g. adding qualifiers) to improve precision. The final search (completed May 31, 2025) retrieved 1468 records. After removing duplicates and screening titles/abstracts for the above themes, the remaining articles were assessed in full text.

4.2. Inclusion and exclusion criteria

To focus our review on relevant, high-quality studies, we defined clear inclusion and exclusion criteria. Each set of criteria was applied during the screening phases. Key categories are summarized below:

- **Relevance to BSGs:**
  - *Included:* Studies explicitly examining business simulation games (BSGs), management simulation games, serious games for business education, or immersive business training simulations. For example, research on a VR-based management simulation or a competitive market business game would qualify.
  - *Excluded:* Studies focused only on gamification elements without a structured business simulation component.
- **Integration of emerging technologies:**
  - *Included:* Studies that implement at least one emerging technology in the context of a BSG. This includes, for example, AI-driven decision support or virtual assistants, VR-based business training, AR scenario overlays, web- or mobile-based simulation platforms, IoT devices for real-time interaction tracking, or blockchain systems for secure student credentialing. (E.g., a simulation using IoT sensors to record student actions, or a blockchain ledger to verify business simulation credits.)
  - *Excluded:* Broad discussions of educational technologies not tied to specific business simulations.
- **Educational context:**
  - *Included:* Research conducted in higher education settings – specifically business schools, management departments, or MBA programs where BSGs are used as instructional tools.
  - *Excluded:* Studies set in K-12 schools, vocational training, corporate training, or other non-university contexts.
- **Study type (empirical evidence):**
  - *Included:* Empirical studies presenting quantitative, qualitative, or mixed-methods data on BSGs. For instance, experimental or survey studies measuring learning outcomes of an AI-enhanced simulation would be included.
  - *Excluded:* Theoretical papers, conceptual articles, opinion pieces, book chapters, or literature reviews, as well as any report lacking sufficient methodological detail (e.g., missing data collection or results).
- **Publication parameters:**
  - *Date:* We conducted an open-ended search without restricting the publication date. The data extraction process began on December 18, 2024, and was completed on May 31, 2025.
  - *Type:* Only peer-reviewed sources (journals and conference proceedings) were included; preprints, dissertations, and other non-peer-reviewed materials were excluded.
  - *Language:* We included only English-language publications, excluding non-English works due to translation limitations.

To manage the large number of articles retrieved (n = 1468), rigorous inclusion and exclusion criteria were applied. Following inclusion and exclusion criteria, 46 studies met the criteria and were included in the final review (Fig. 1).

4.3. Quality assessment

The methodological quality of the 46 included studies was appraised with criteria (Tables 1 and 2) adapted from Kmet et al. [40]. Separate check-lists were applied to quantitative (including experimental / quasi-experimental designs), qualitative, and mixed-methods investigations. Each item was scored 0 = No or 1 = Yes; non-applicable items were marked N/A. A study's quality score was the sum of 'Yes' responses divided by the maximum applicable items, yielding a proportion between 0 and 1. No study was excluded on quality grounds, but the synthesis weighted evidence according to these scores.

Across the 46 studies audited—34 quantitative or quasi-experimental, 11 mixed-methods, and one qualitative—the overall methodological quality was strong. Quantitative scores spanned 0.70–1.00 (M = 0.93, SD = 0.06); 28 of the 34 quantitative papers (≈82 %) exceeded the strict benchmark of 0.75, while the remaining six lay between the lenient (0.55) and strict thresholds, and none fell below 0.55. The single qualitative study achieved a score of 0.88. Within the mixed-methods set, quantitative components mirrored this distribution, whereas qualitative components ranged 0.80–0.95 (M = 0.88). Adhering to Kmet et al.'s guidance—scores ≥0.55 as a lenient minimum and ≥0.75 as a strict minimum—all studies comfortably met at least the lenient criterion; lower-scoring papers were proportionally down-weighted in the ensuing synthesis.

4.4. Inter-rater reliability

Two researchers independently applied the Kmet et al. [40] check-lists. After a joint calibration exercise on five pilot papers, they double-coded a stratified random 20 % subsample (n = 9) that contained proportional numbers of quantitative, qualitative, and mixed-methods studies. Cohen's κ values for individual check-list items ranged from 0.74 to 0.92. The overall study-level agreement was κ = 0.84 (95 % CI 0.78 – 0.90), classified as substantial. Raw percentage agreement averaged 92 %. Items with κ < 0.60 (e.g., the randomisation procedure in quasi-experimental designs) were discussed, code-book wording was refined, and all discrepancies were resolved by consensus before final scoring. A sensitivity check on a fresh 10 % verification sample produced a comparable κ = 0.82, supporting the stability of the ratings.

4.5. Data analysis

We analyzed the included studies using thematic synthesis, following established qualitative synthesis methods. Referring the methodology and guidance of the previous studies [68],[2], the procedure in this review was carried out in three phases. First, all authors thoroughly read each paper and coded relevant findings (e.g. on technology use or learning outcomes). Codes were iteratively refined as more studies were reviewed. In a second round, authors compared and consolidated codes, grouping them into broader categories. Finally, these categories were examined to identify analytical themes reflecting core insights across studies. Throughout this process, we regularly reviewed and revised codes to ensure consistency and credibility. The end result was a set of coherent themes describing how emerging technologies integrate with BSGs in business education.

5. Results

5.1. Methodological trends and research context in the reviewed articles

The review analyzed 46 studies on the role of different emerging technologies in business simulation games (Table 3). To capture the

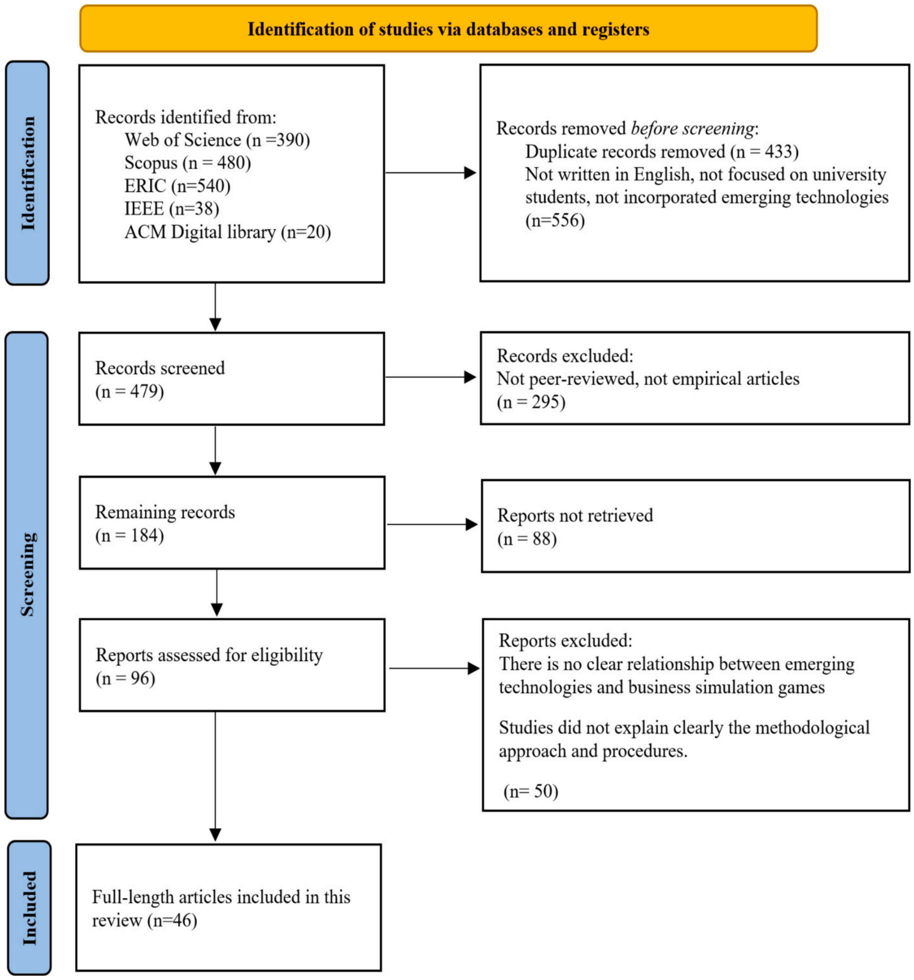

Fig. 1. PRISMA flow chart.

Table 1
Criteria for quantitative studies.

No.	Criterion
1.	Is the research question or objective clearly stated and well explained?
2.	Is the study design clearly identified and suitable for the research purpose?
3.	Are the methods for selecting participants or data sources properly described and appropriate?
4.	Are the characteristics of the participants (and comparison group, if any) clearly outlined?
5.	If the study involves an intervention, is the process of random assignment explained?
6.	In interventional studies, is there any mention of whether the investigators were blinded?
7.	In interventional studies, is blinding of participants addressed?
8.	Are the outcome and exposure measures clearly defined, reliable, and free from measurement error?
9.	Is the method of assessment explained?
10.	Is the sample size adequate to support reliable conclusions?
11.	Are the data-analysis techniques properly explained and appropriate for the study design?
12.	Are measures of variability (e.g., confidence intervals or SDs) provided for the main findings?
13.	Have the researchers considered and controlled for potential confounding variables?
14.	Are the findings presented in sufficient detail for the reader to understand the results?
15.	Do the conclusions accurately reflect and follow from the results?

Table 2
Criteria for qualitative studies.

No.	Criterion
1	Is the research question or objective clearly described?
2	Is the study design clearly outlined and appropriate?
3	Is the context of the study well presented?
4	Is there a clear connection to a theoretical framework or broader literature?
5	Is the sampling strategy explained, appropriate, and well justified?
6	Are the data-collection methods clearly outlined and applied systematically?
7	Is the data-analysis process clearly described and methodologically coherent?
8	Are verification procedures used to enhance the credibility of the findings?
9	Is evidence of reflexivity (e.g., researcher influence) provided?
10	Does the account demonstrate reflexivity (e.g., positionality, influence)?

educational and practical impacts of simulation games, the studies were categorized based on their research designs and data analysis approaches (More broadly in Appendix B). In terms of research design, the most common approach was quasi-experimental (n = 14), followed by experimental designs (n = 6), both typically employing pre-test and post-test measures. A smaller number of studies utilized case study or exploratory designs (n = 2). Regarding data analysis methods, the majority of studies employed quantitative approaches (n = 13), using tools such as surveys, questionnaires, and descriptive statistics. Mixed-method approaches were also prevalent (n = 12), integrating both qualitative and quantitative data. In contrast, qualitative-only approaches were the least represented, with just one study (n = 1) using interviews or narrative analysis to explore the phenomenon in depth.

The regional distribution of these studies reflected a wide range of

Table 3
Reviewed study objectives and main results.

ID	Article	Research objectives	Main findings
1.	Ahmed et al. [3]	To explore the role of AI-supported serious simulation games in enhancing students' understanding and application of urban planning concepts.	AI-supported serious simulation games significantly enhance students' understanding of urban planning concepts and improve their critical thinking skills through interactive learning experiences. Additionally, these games foster students' engagement and collaboration opportunities.
2.	Alnuaim [4]	To assess the effectiveness and learner acceptance of gamification in a digital literacy course at the undergraduate level using a randomized controlled trial.	The study found that gamification tools significantly improved academic performance in digital literacy among undergraduate students. However, prior experience with gamification did not significantly influence their acceptance of this approach in the course, highlighting its potential benefits in education.
3.	Bamufleh et al. [7]	To investigate students' acceptance of simulation games in management education and the factors influencing their intention to use such tools.	The study explores factors influencing students' acceptance of simulation games in management courses, identifying performance expectancy, effort expectancy, social influence, and facilitating conditions as key determinants.
4.	Buzady & Almeida [9]	To evaluate the effectiveness of the FLIGBY game in enhancing motivation and competencies related to entrepreneurship education.	FLIGBY is a serious game designed to enhance entrepreneurship skills by immersing students in real business challenges. It promotes the development of management, leadership, and soft skills, while incorporating assessment elements to monitor and analyze student performance effectively.
5.	Chatpinyakoop et al. [10]	To assess how online simulation-based learning impacts students' change management skills in the context of corporate sustainability.	The study found significant improvements in students' execution of theory-informed change strategies through a business simulation focused on sustainability practices. Learning module effectively enhanced students' management skills related to corporate sustainability.
6.	Chen et al. [11]	To investigate the impact of an AI module on students' entrepreneurial attitudes in a BSG.	AI realistically simulates opponents, provides accurate feedback on failures, and enriches the game environment; this significantly increases students' positive attitudes towards entrepreneurship.
7.	Chen et al. [12]	To measure the effects of mobile business games on entrepreneurial skills, self-confidence, and intention.	The sense of "flow" generated by the game enhances students' entrepreneurial attitudes and self-confidence, but entrepreneurial intention does not change; this indicates a mismatch between attitude and intention.
8.	Davis et al. [13]	To examine the impact of team goal orientation (eagerness to learn or fear of appearing incompetent) on performance under complex simulation conditions.	Learning-oriented teams outperform their competitors; fear of appearing incompetent leads to risk aversion, which weakens performance. Realistic simulation tools play an important role in uncovering this process.
9.	Endress et al. [16]	To assess the ability of a digital serious game to increase student engagement and active learning in virtual business classes.	Simple yet goal-oriented games significantly increase student motivation and learning outcomes by enhancing group decision-making, interaction, and knowledge application. Findings indicated that a simple game can effectively promote participation and learning among students.
10.	Enang & Omeihe [15]	To design, develop, and evaluate a real-time, adaptive AI-driven business simulation platform that addresses the limitations of static business simulations by incorporating machine learning, NLP, and knowledge graph technologies. The study aimed to enhance strategic decision-making, adaptive reasoning, and knowledge transfer through dynamic, evolving scenarios.	The results indicated that the system significantly strengthens users' ability to make strategic decisions, facilitates the sharing and application of knowledge, and promotes flexible, context-aware thinking. This study paves the way for future business education and strategic planning platforms that more accurately capture the dynamic and intricate nature of real-world business environments.
11.	Fuchsberger [19]	To examine the impact of the artificial intelligence expert system "Eddie" on students' decision-making skills in a business game setting.	"Eddie" enhanced students' decision-making in marketing, product development, competitor analysis, and salary policy; participants highlighted the tool as particularly useful in the areas of market analysis,

(continued on next page)

Table 3 (*continued*)

ID	Article	Research objectives	Main findings
			logistics, and process optimization.
12.	Gatti et al. [21]	To explore how sustainability-focused gamification can influence students' learning outcomes in business education.	The study found that the game Napuro effectively enhances students' understanding of sustainability issues and positively influences their attitudes and intentions toward sustainable behavior. It demonstrates that simulation-based action learning can lead to both cognitive and emotional learning outcomes, potentially strengthening critical thinking skills. Importantly, students with higher motivation and interest in sustainability prior to the game showed greater gains in knowledge and attitude shifts, highlighting motivation as a key factor in learning effectiveness.
13.	Gawel et al. [22]	To implement sustainability principles into virtual simulation games and examine their effectiveness in business education.	Virtual BSG was effective in integrating sustainability into business education. The study identified key design principles for sustainability-focused VSGs, including real-world relevance, interactivity, and feedback mechanisms. Students developed systems thinking and ethical decision-making skills related to sustainability. Challenges included technical limitations and the need for instructor training to maximize effectiveness.
14.	Gironacci [23]	To evaluate the usability and effectiveness of an XR simulator supported by content-based scenario recommendation in management training.	Personalized XR scenarios enhance decision-making skills and adaptability; intelligent recommendation mechanisms increase motivation by adapting to the learner's needs.
15.	Grivokostopoulou et al. [24]	To study the long-term impact of gamified entrepreneurship training on students' entrepreneurial intentions and skills..	Gamified elements in entrepreneurship education such as scores, badges, and leaderboards enhance students' motivation, creativity, and risk assessment skills. The study suggested that long-term gamification leads to deeper learning compared to short-term interventions.
16.	Hoang et al. [28]	To evaluate the combined effects of human-system interaction, social expectations, and self-confidence on Vietnamese students' entrepreneurial intentions.	Self-efficacy fully mediates the relationship between human-system interaction and entrepreneurial intention; social expectations enhance this effect. Thus, simulation games increase both self-confidence and entrepreneurial intention in the presence of social support.
17.	Hsu & Wu [29]	To examine the impact of business simulation games on student engagement and Higher-Order Thinking Skills (HOTS) in a flipped classroom model.	BSGs increase student engagement and higher order thinking skills in the flipped classroom; behavioral engagement enhances problem solving, emotional engagement enhances all forms of HOTS, and cognitive engagement enhances critical thinking and creativity.
18.	Huang et al. [30]	To investigate the relationship between learner engagement in business simulation games and the development of higher-order thinking skills.	The majority of students achieved the learning outcomes, developing key skills in analyzing business challenges, defining system requirements, and designing solutions. Participants described the simulation as both realistic and stimulating, highlighting its role in building practical knowledge and increasing their enthusiasm for learning.
19.	Humpherys et al. [31]	To foster tacit knowledge and problem-solving abilities in students through experiential learning via a role-play business simulation.	Business simulation games, when combined with project-based learning (PBL) in flipped classrooms, actively boost student learning outcomes. This approach develops critical thinking and problem-solving skills, and enhances creativity.
20.	Isabelle [32]	To evaluate how gamifying an entrepreneurship course through real online business creation influences engagement and learning.	Live leaderboards enhance competition, significantly enhance student experience, engagement, and entrepreneurial confidence; working with real money makes learning concrete.
21.	Jagger et al. [33]	To measure the effectiveness of a fully immersive 3D game called "Marketing Mayhem" in teaching business ethics.	The game personalizes ethical dilemmas, making learning more immersive and engaging for students; however, further research is needed to explore its suitability for different learning styles.

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Table 3 (*continued*)

ID	Article	Research objectives	Main findings
22.	Jiang et al. [34]	To compare the teaching effectiveness of a VR-based experiential learning environment for international negotiations with traditional methods.	This study developed a VR learning model for business negotiation using 3Ds Max and VR-platform that improved memory retention, comprehension, motivation, and educational outcomes compared to traditional teaching methods.
23.	Kong & Feng [38]	To analyze student perceptions of game-like and instructional VR systems in blended wine education and identify optimal VR design criteria.	Findings suggest effective VR integration requires improved UI/UX elements like pedagogical labelling, sensory integration, and language skill development. The research concludes that properly designed VR can provide valuable feedback and create immersive blended learning experiences that enhance wine education effectiveness.
24.	Korre [39]	To assess usability outcomes of humanoid conversational agents in mobile serious games based on human-likeness and agent number.	Results showed that users significantly preferred HECAs over text-based agents, with a large effect size (d = 1.01). Participants found HECAs more appealing due to their human-like characteristics, highlighting their potential to enhance user interaction and usability in serious games.
25.	Krajčovič et al. [41]	To evaluate how VR-based immersive simulations enhance learning by linking educational content to practical application.	The study showed that VR increased student engagement and motivation by creating an immersive and interactive learning environment. It helped students better understand and apply theoretical concepts in practical contexts. Students also reported that VR made learning more effective, realistic, and closely connected to real-world professional situations.
26.	Lau & Lee [43]	To examine how a 360-degree stereoscopic VR environment influences learning outcomes in business education.	The study found that a stereoscopic 360-degree learning environment significantly improved business students' learning achievement compared to traditional classroom methods. However, it was less effective in fostering students' motivation for independent learning and self-evaluation.
27.	Lean et al. [44]	To study how cognitive realism and decision-making strategies interact to affect student performance in simulation-based learning.	The study found three different ways players perform in a business simulation game. High performers always use flexible decision-making. When the game feels realistic, being more focused leads to better results. But if the game doesn't feel realistic, players tend to try out different strategies, which often leads to poor performance. Also, trying random strategies while seeing the game as realistic does not lead to high performance.
28.	Lee et al. [45]	To evaluate Google Cardboard's effectiveness as a VR content delivery method in business classrooms.	The study found that Google Cardboard VR increased student enjoyment and interest compared to traditional flat-screen (FS) formats, offering a more engaging and immersive learning experience. However, VR did not outperform FS in terms of content delivery, novelty, or clarity. While it enhances motivation and simulates real-world scenarios, its educational impact is limited, and practical challenges like resource demands and training remain.
29.	Lin & Wang [47]	To examine the effects of VR- and animation-supported gamified learning on achievement and behavioral intentions in retail education.	Results showed that cognitive and affective responses positively influenced student confidence and satisfaction, with satisfaction significantly affecting behavioral intention and learning achievement. However, confidence did not significantly impact behavioral intention or learning achievement after four weeks, suggesting that immediate confidence may not translate to sustained learning outcomes.
30.	Milić et al. [48]	To design and evaluate a hybrid fuzzy–crisp AI system for teaching key performance indicators, focusing on its effectiveness in improving KPI understanding, retention, and usability.	This study referred to the a hybrid fuzzy-crisp AI system to teach key performance indicators to management students at the University of Belgrade. This system offered interactive, intuitive, and adaptive simulations. The hybrid fuzzy-crisp AI approach via ESPRIT enhances learning by simulating realistic business

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Table 3 (*continued*)

ID	Article	Research objectives	Main findings
			scenarios with an interactive, user-friendly interface. Students using ESPRIT showed higher performance and better knowledge retention than those in traditional classes.
31.	Musshoff & Hirschauer [49]	To test if simulation games can evaluate behavioral effects of agricultural policy instruments aimed at reducing nitrogen leaching.	The study found that formal regulations with sanctions were more effective in reducing nitrogen input among participants simulating farmers, compared to voluntary schemes with incentives. The findings emphasize the role of nonmonetary motivations and bounded rationality in shaping responses to environmental policies. However, external validity of the study is limited, as the participants were students whose decisions may not fully reflect those of real farmers, and the simulation did not capture the full complexity of real-world farming environments.
32.	Newbery et al. [50]	To determine whether participation in serious games affects students' entrepreneurial intentions.	The serious game (SimVenture TM) significantly decreased entrepreneurial intention among participants, suggesting it exposed the challenges and complexities of entrepreneurship in a discouraging way. Gender and role model effects were also significant factors influencing entrepreneurial intentions.
33.	Nikolaeva et al. [51]	To develop pre-service restaurant managers' English speech interaction skills through simulations and assess the role of GPTChat.	Pre-service restaurant managers who combined teacher consultation, ChatGPT, and business simulation games showed significantly higher speech interaction and decision-making skills than those who used ChatGPT alone. Business simulation games improved understanding of professional communication contexts, while GPTChat effectively supported scenario creation and decision-making analysis.
34.	Pacheco-Velázquez et al. [53]	To explore how serious games enhance self-directed learning skills and propose a framework for measuring this growth.	The study showed that business simulation positively influenced self-directed learning (SDL) readiness by enhancing critical thinking, reasoning, analysis, and decision-making. Five key SDL factors were identified: self-management, openness to learning, initiative, self-concept as a learner, and desire for learning.
35.	Pérez-Pérez et al. [57]	To assess the impact of a serious business game on students' entrepreneurial intentions and analyze moderating personal factors.	The study shows that BSG helps students understand the challenges of entrepreneurship. Gender does not affect how these games influence entrepreneurial intention (EI). Interestingly, students with higher academic performance tend to show lower EI after playing the game, while those with no close ties to entrepreneurship experience a decrease in their EI.
36.	Ronaghi & Forouharfar [60]	To investigate how virtual reality-based simulated experiences influence students' entrepreneurial intentions in entrepreneurship education.	The research found that VR-based entrepreneurship training significantly increased entrepreneurial intention among Iranian university students compared to both traditional teaching methods and a control group. The study showed VR technology positively affected students' personal attitudes and perceived behavioral control toward entrepreneurship, though neither training approach impacted subjective norms.
37.	Sari et al. [61]	To evaluate the effectiveness of VR-based business ethics teaching across various learning styles.	VR offers flexible learning unconstrained by time, distance, or physical space. Its immersive, interactive, and imaginative nature enhances cognitive performance in fields like engineering, military training, and surgical robotics. VR improves learning outcomes by simulating real-world scenarios, particularly in business ethics, where students safely explore dilemmas, observe consequences, and practice decision-

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Table 3 (*continued*)

ID	Article	Research objectives	Main findings
			making. This experiential approach builds long-term skills and prepares learners for a future blending virtual and physical environment.
38.	Sheikh et al. [63]	To assess whether business simulation workshops enhance students' 21st-century skills, such as entrepreneurship and employability.	The simulation significantly improved entrepreneurial and employable skills but has minimal impact on sustainability skills. Additionally, factors such as gender and expertise showed little influence on the outcomes.
39.	Silitonga et al. [64]	To analyze the effect of business simulation games on students' entrepreneurial intentions and competencies through experimental research.	This study focuses on the effectiveness of BSG in enhancing entrepreneurial intentions and competencies among students through hands-on experience. Findings show that the proposed BSG improves cognitive and non-cognitive entrepreneurial competencies, and thus, enhances students' intention to start a new business.
40.	Stamatakis et al. [65]	To explore how blockchain technology can enhance user engagement and learning in serious games through decentralized mechanics.	This study explores the use of blockchain technology in a serious card game focused on renewable energy and sustainable resource management. It examines how blockchain features—such as virtual asset ownership, transparent transactions, trustless game mechanics, and cross-platform asset reuse—enhance user interaction and ownership in gaming. The game serves both as entertainment and an educational tool, demonstrating blockchain's potential to create secure, interactive, and awareness-raising virtual environments.
41.	Sung et al. [66]	To assess the effectiveness of VR in marketing education using both self-report and psychophysiological measurements.	The study found that VR increased immersion and learning attitude but not enjoyment, while reducing test performance versus traditional video. VR only improved learning attitude through immersion, not enjoyment or knowledge retention.
42.	Wendt et al. [70]	To design a blockchain-based token trading game that teaches students supply chain management and market interaction.	The blockchain-enabled token trading game provides participants with practical experience in using blockchain technology for supply chain management, allowing them to understand the complexities of trading supplier capacity in response to uncertain customer demand. The game serves as an interactive pedagogical tool that helps students and executives grasp the challenges of supply chain management and the effectiveness of virtual markets in reconciling supply and demand, ultimately aiming to enhance overall supply chain performance.
43.	Yang et al. [73]	To investigate the role of student self-efficacy and instructional methods on learning outcomes in mobile business simulation games.	The study explores how virtual simulation games impact student engagement and entrepreneurial skill development, highlighting that teamwork and self-efficacy significantly influence skill development, while certain game design experiences do not positively affect engagement, despite their importance in learning outcomes.
44.	Yang et al. [74]	To analyze how simulation game experiences influence engagement and entrepreneurial skill development among students.	The study found that student self-efficacy significantly influenced learning outcomes, with hierarchical teaching methods enhancing self-efficacy and engagement more than general methods. Self-efficacy directly affected learning outcomes in hierarchical learning settings, while it mediated engagement in traditional learning settings.
45.	Zhang [76]	To determine whether login frequency and consistency are valid indicators of student engagement in business simulations.	In this study, the login consistency of participants showed a stronger link to game involvement than login frequency, making it a better indicator of engagement. Research and teaching implications are discussed.
46.	Zulfiqar et al. [78]	To evaluate the influence of business simulation games on entrepreneurial intentions among students during COVID-19 using TAM.	The study investigates the effectiveness of online management simulation games in improving students' learning performance and entrepreneurial intentions. These games

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Table 3 (*continued*)

ID	Article	Research objectives	Main findings
			were compared to traditional theoretical online lectures to assess their impact on student engagement and learning outcomes

cultural and geographical contexts. China emerged as the most frequently studied region, with ten studies (n = 10). The United States and UK was the second most represented countries, with four studies (n = 4), respectively. Additionally, other regions including Saudi Arabia (n = 3), Canada (n = 2), Australia (n = 2), Germany (n = 2), Greece (n = 2), Thailand (n = 2). In comparison with these regions, other regions, including Austria, Switzerland, Poland, Portugal, Slovakia, Serbia, Spanish, Mexico, Indonesia, Pakistan, Vietnam, Ukraine, Iraq, Iran were represented with one study (n = 1). In addition to these studies, only Lean et al. [44] provided a comparative analysis of the effectiveness of simulation games across multiple regions, focusing specifically on the United Kingdom and China (Hong Kong) (Fig. 2).

The technological foundations of the reviewed business simulation games varied widely, reflecting the diversity of emerging technologies being integrated into educational and business contexts. The most commonly used platforms were web-based simulation games (n = 18) referring to the open-source platforms, followed by VR-based environments (n = 13), which often aimed to enhance immersion and experiential learning. AI-powered simulation systems were represented in five studies (n = 5), including two that explicitly employed generative AI chatbots (n = 2) to support learner interaction and decision-making. One of the study integrated XR simulator with AI-powered recommendation mechanisms [23]. In addition, mobile-based (n = 2) and blockchain-integrated simulation games (n = 2) were explored, though these technologies were less frequently studied.

5.2. Educational benefits of emerging technologies with BSGs

Although instructors used business simulation games in the previous years for the simpler decision-making processes, the integration of innovative technologies into these games became game changers by the time. Recent development of technology tools (artificial intelligence, generative AI chatbots, virtual reality, blockchain-integrated simulations as well as mobile-based and web-based simulations) enable students to solve more complex problems in an interactive learning environment. We mapped out the benefits of the different technology types based on our limited reviewed studies. The following six themes (Table 4) summarize the key educational benefits of integrating emerging technologies into business simulation games.

The integration of AI systems offers different benefits for simulated

Table 4
Benefits of integrating emerging technologies into BSGs.

Theme	Technology types	Benefits
Personalized and adaptive learning	AI, Generative AI (e.g., ChatGPT) chatbots, mobile virtual assistants	Real-time personalized feedback Adaptive challenges based on user behavior Supports student-centered learning
Development of critical and analytical thinking skills	AI-powered simulations, virtual reality, Generative AI (e.g., ChatGPT) chatbots	Enhances decision-making in complex scenarios Strengthens analytical reasoning Supports experiential learning
Immersive and realistic practice environments	Virtual reality, humanoid conversational agents	Simulates real-world business settings Enables safe experimentation Develops business acumen and adaptability
Fostering entrepreneurial mindset and confidence	Mobile simulations, AI systems, VR	Encourages initiative and innovation Boosts self-efficacy in business decision-making Improves risk assessment and planning
Integration of emerging digital competencies	Blockchain, gamification, web-based simulations	Introduces decentralized systems and digital credentials Enhances understanding of token-based economies Builds digital finance literacy
Improvement in engagement, performance and knowledge retention	Mobile simulations, VR, AI feedback systems, web-based simulations	Increases motivation and academic performance Supports real-time learner feedback and problem-solving Improves long-term knowledge retention

Fig. 2. Regional distribution of studies on simulation games.

learning in business schools. Recent studies [3,11,15,19] explain that the AI-powered simulation games provide users with real-time personalized feedback and offer tasks that are tailored to their skill. These personalized approaches increase entrepreneurial mindset in students. Moreover, AI-powered business simulations develop students' analytical thinking by replicating real-life problems. Fuchsberger [19] shows that AI helps students build more accurate strategies and reduces logical errors significantly. This approach effectively supports experiential learning. Likewise, the simulation game developed by Enang and Omeihe [15] with the integration of AI was highly valuable as it was scalable for individual or institutional training. As well as, the AI system continuously learn from users' behavior, which this process mimics real-world volatility in business education. Buzady and Almeida [9] conducted a mixed methods study in Portugal with 51 students, demonstrating that the AI-based serious game FLIGBY effectively enhances entrepreneurship, management, leadership, and soft skills through immersive business challenges and integrated performance assessment.

In addition, if business simulation games are incorporated with artificial intelligence chatbots like ChatGPT, users can improve their critical thinking, complex problem-solving and decision-making skills. For example, Nikolaeva et al.'s [51] mixed-methods research showed that business simulation games empowered by ChatGPT enhanced speech interaction and decision-making skills in conflict and non-conflict scenarios for 44 pre-service restaurant managers. Generative AI chatbot effectively generated professional scenarios and analyzed decision-making algorithms, serving as a valuable supplemental tool in experiential learning. Similarly, Korre [39] showed that humanoid conversational agents (HECAs) in mobile games outperformed text-based agents in satisfaction and usability, integrating natural spoken interaction and affective elements for immersive learning.

As for virtual reality technologies, these technologies allow students to apply the knowledge they have learned in a real business environment. First, VR simulations help students develop workplace skills through practical experience. Jiang et al. [34] and Krajčovič et al. [41] found these technologies improve conversation abilities while speeding up decision-making. Students gather information in real time to assess risks, becoming more adaptable thinkers. Second, immersive technologies significantly enhance students' critical thinking and business acumen in several key ways. Huang et al. [30] and Hsu and Wu [29] documented how safe virtual environments strengthen analytical reasoning. Further research by Chen et al. [11] and Hoang et al. [28] revealed improvements in entrepreneurial mindset, especially in risk assessment and business planning.

In parallel with these technologies, few studies conducted the role of blockchain technologies in educational gaming. The results from the studies showed that blockchain technology-integrated BSGs foster user agency and strategic interaction [65,70]. In this case, students can develop the knowledge for trust to their colleagues in the digital world and managing the digital resources. More broadly, Wendt et al. [70] developed a supply chain simulation where students used token-based trading to explore how decentralized economic systems operate. Stamatakis et al. [65] have used blockchain technology to securely implement tasks such as tracking academic achievement, managing virtual classroom economies, and issuing digital certificates. However, its practical utility and pedagogical relevance are yet to be fully established.

The increased use of mobile technologies has made business simulation games (BSGs) more accessible to students. Chen et al. [12] found that mobile BSGs encouraged entrepreneurial thinking and boosted students' confidence in their abilities. Similarly, Yang et al. [73] examined the impact of integrating structured teaching strategies into these games, noting improvements in engagement and academic performance. In comparison with this research, Korre [39] focused on human-like virtual assistants in mobile simulations, which appeared to support a more interactive and student-centered learning environment.

The educational value of BSGs lies in their ability to develop essential cognitive and professional skills. Research shows that these games help students strengthen problem analysis [31], enhance critical thinking [3,21], and gain a clearer understanding of key business areas such as market analysis [19]. They also support the development of decision-making skills, particularly in uncertain or complex situations [13]. Moreover, immersive tools and AI-powered applications—such as virtual assistants and chatbots—have been linked to improved problem-solving and learner support [19,31,53]. In particular, AI and virtual reality technologies provide real-time feedback and adaptive learning experiences that foster continuous development [11,13,22]. These tools also promote better knowledge retention over time [48].

5.3. Methodological and pedagogical limitations

The review explored the limitations of emerging technology-integrated BSGs focusing on technical and pedagogical aspects (Table 5). The key limitations include difficulty in knowledge integration, limited effectiveness in some parts of learning performances, UI/UX design shortcomings, the gap between external validity and real-world scenarios, and the limited capacity for fostering independent learning and self-assessment.

While AI-powered business simulation games (BSGs) have shown promise in enhancing decision-making, entrepreneurial skills, and adaptability (e.g., [15,51], several limitations have been identified. One key concern is the limited scope of AI feedback. For instance, Sheikh et al. [63] found that while AI simulations improved employability and entrepreneurial skills, they had minimal impact on sustainability-related skills, indicating the insufficient breadth of AI systems in addressing

Table 5
Limitations of emerging technology-integrated business simulation games.

Technology type	Key limitations	Studies
AI-based (Machine learning, conversational agents, chatbots)	• Limited feedback scope (e.g., on sustainability) • May hinder independent thinking • High technical complexity • Instructor dependency	Sheikh et al. [63], Fuchsberger [19],Milić et al. [48]
VR-based (semi-immersive – 3D animated; fully immersive –business training simulators; mobile VR – 360-degree videos)	• Perceived lack of realism reduces strategic engagement • May discourage entrepreneurship • Limited external validity • Engagement may not suit all learners	Lee et al. [45],Sung et al. [66],Ronaghi & Forouharfar [60], Grivokostopoulou et al. [24],Krajčovič et al. [41]
Web-based	• Perceived lack of realism reduces strategic engagement • May discourage entrepreneurship • Limited external validity • Engagement may not suit all learners	Lean et al. [44],Newbery et al. [50],Musshoff & Hirschauer [49],Jagger et al. [33]
Mobile-based	• Improved attitudes but not intentions • Requires structured teaching support • May lack depth and pedagogical guidance	Chen et al. [12],Yang et al. [73],Korre [39]
Blockchain-based	• High abstraction and complexity • Steep learning curve • Requires scaffolding for comprehension and application	Stamatakis et al. [65], Wendt et al. [70]

complex, interdisciplinary learning outcomes. Similarly, Fuchsberger [19] revealed that while students trusted the AI advisor, "Eddie," their reliance on AI guidance might hinder independent strategic thinking, particularly in later simulation rounds. Milić et al. [48] noted that although the hybrid fuzzy-crisp AI system offered adaptive experiences, technical sophistication required substantial instructor facilitation, revealing a steep learning curve and high demand on teacher training and technical infrastructure.

Research shows VR technologies create immersive environments that boost student motivation in complex subjects like sustainability and ethics [22,34]. However, several limitations exist. Technical complexity and equipment requirements create barriers, especially in resource-limited institutions [24,41]. While VR enhances enjoyment, it does not necessarily improve content delivery compared to traditional formats and demands extensive instructor support [45]. Moreover, Sung et al. [66] found VR actually decreased test performance despite increased immersion, possibly due to cognitive overload. Additionally, Ronaghi and Forouharfar [60] demonstrated the limited impact of VR-integrated interventions on subjective norms.

On the other hand, the review also explained the challenges of virtual reality-based BSGs regarding their user interface and experience design. Kong and Feng [38] proved the challenges through insufficient pedagogical labelling and lacked sensory integration for increased immersion. These results further demonstrate that games alone cannot provide pedagogical guidance. Furthermore, it should be noted that the lack of immersive elements in students' ability to obtain clear instruction results in a decrease in their interest or entrepreneurship pursuits in the game. Interestingly, Musshoff and Hirschauer [49] noted that emerging simulation games are still designed with limited external validity as in the traditional BSGs, which is a barrier to training students based on real business situations.

Web-based BSGs are widely accessible and have demonstrated success in improving self-directed learning, critical thinking, and entrepreneurial competencies (e.g., [53,64]. However, some studies pointed out limitations in realism and learner engagement over time. For example, Lean et al. [44] found that lack of perceived realism in web-based simulations caused students to adopt random or less-strategic behaviours, which negatively affected performance. Newbery et al. [50] raised concerns that simulations could sometimes discourage entrepreneurship by overemphasizing complexity and risk, reducing entrepreneurial intention. Additionally, Musshoff & Hirschauer [49] argued that student participants may not accurately reflect real-world actors, thus questioning the external validity of web-based BSG findings in policy-related simulations like farming. Moreover, Jagger et al. [33] emphasized that while 3D web-based games enhanced engagement, their suitability for diverse learning styles remained unclear.

Mobile business simulations are appreciated for their portability and user engagement. Studies like Chen et al. [12] and Yang et al. [73] highlighted their impact on self-efficacy and entrepreneurial attitudes. However, limitations exist in terms of depth of learning and motivation. Chen et al. [12] observed that although mobile games improved entrepreneurial attitudes and self-efficacy, they did not lead to changes in entrepreneurial intentions, indicating a gap between attitudinal change and behavioral intent. Yang et al. [73] found that hierarchical teaching methods were needed to supplement mobile BSGs to maximize learning outcomes, suggesting that mobile games alone might lack pedagogical structure to support diverse learner needs.

Blockchain-enabled simulations introduce novel features such as asset ownership, trustless mechanics, and transparent transactions; however, complexity and user adaptability were notable limitations. Stamatakis et al. [65] pointed out that despite fostering interaction and awareness, blockchain mechanics may be too abstract or technical for some learners, requiring prior knowledge or additional scaffolding. Wendt et al. [70] added that while blockchain simulations enhanced understanding of supply chain complexities, practical application remained challenging due to the steep learning curve associated with decentralized technologies.

Moreover, several studies [21,44,57,76] also suggested that innovative technologies are not particularly conducive to low-achieving students. The result indeed shows the educational inequities of these games. The research carried out by Lau and Lee [43] explored the impact of these emerging games for independent learning and self-evaluation. They identified that the new BSGs can be less effective for these approaches. The rationale behind this finding could be the lack of personalization and motivational features in BSGs.

5.4. Cognitive and contextual design factors

Recent innovations in artificial intelligence, generative artificial intelligence, virtual and extended reality technologies, mobile applications, blockchain technologies, and web environments have significantly expanded the pedagogical potential of business simulation games (BSGs); however, learner achievement still depends primarily on the the different factors in relation to the design of these games – how the features are configured.

Realism and scenario design. Authentic scenarios motivate strategic, goal-oriented behaviour. High perceived realism has been linked to more focused decision-making [44], whereas low realism encourages trial-and-error approaches with weaker outcomes. VR, XR, and blockchain settings replicate real-world complexity, fostering negotiation skills and ethical reasoning [34,61,65,70]. AI-infused games augment realism through dynamic competitor modelling and situational feedback [11].

Technology-specific interaction and usability. AI-based BSGs perform well when they provide contextual feedback and personalized support. Highly realistic dialogue agents [39] and adaptive simulation platforms [15]) create both interest and trust in learners. In VR games, immersive environments, comfortable controls, and ergonomic devices [38,45] directly increase motivation. Mobile BSGs emphasize comfort and a sense of "flow"; hierarchical learning design [73] and mobile-specific engagement methods [12] enhance self-confidence and cognitive focus. Blockchain-integrated games, on the other hand, support player agency and strategic behavior through transparent mechanisms and verifiable asset ownership [65,70].

Instructor support and training. Even technically advanced platforms can lose their pedagogical impact without teacher support. Research shows that prior training of teachers in VR, AI, GenAI and blockchain environments is essential for effectively guiding students through complex scenarios [22,47].

Pedagogical design and instructional strategies. The instructional framework must complement the technology. Project-based learning, flipped classrooms, and guided reflection have consistently amplified the educational value of BSGs [10,29]. Feedback loops and iterative practice nurture critical thinking and sustainability-oriented reasoning [16,21]. Additionally, culturally responsive scenarios—such as localised entrepreneurship challenges [63,64]—heighten relevance and learner receptivity, underscoring the merit of streamlined, customisable game designs.

Contextual and institutional influences. Beyond classroom practice, cultural norms and regulatory landscapes modulate how learners interpret simulations. Adaptive AI frameworks, for instance, can mirror regional market dynamics to ensure contextual authenticity [15]. Regulatory considerations, especially in sustainability-focused games, shape permissible decision spaces [49]. Hence, institutional alignment remains critical for maximising the benefits of BSGs.

Motivational and psychological factors. Regardless of platform, motivation remains pivotal. Gamified elements—points, badges, leaderboards—drive participation in web-based and mobile contexts [24,32]. Trust in AI agents correlates with stronger performance in AI-enhanced settings [19], while VR immersion elevates intrinsic motivation even if test scores do not always follow suit [66].

Cognitive and affective engagement. Sustained learning benefits arise when cognitive, behavioural, and emotional engagement converge. Goal orientation, effective teamwork, and robust feedback systems in web-based games promote knowledge retention [13,53,74]. Personalisation and real-time decision support deepen engagement in AI- and VR-based environments [23,48]. Moreover, consistent participation patterns [76] and positive emotional involvement [30] predict higher-order thinking. Adaptive feedback and nuanced scenario complexity further encourage strategic flexibility [15,63].

6. Discussion

This review synthesizes findings from 46 empirical studies on the role of emerging technologies in business simulation games (BSGs). Three key themes emerge: the benefits of BSGs with emerging technologies, the factors promoting active learning in these games, and the limitations of technology integration. Overall, BSGs are beneficial in preparing future business leaders when incorporating technologies such as AI, Generative AI chatbots, VR, and mobile and web-based simulation games along with blockchain technologies, though challenges related to technology integration and user factors remain. Although our initial search strategy explicitly included terms such as “blockchain,” “IoT,” “AR,” and “XR” to ensure comprehensive coverage, only a small number of studies were ultimately included that addressed these technologies. As such, their discussion in this section is framed within the context of future research potential and theoretical relevance, rather than empirical frequency.

6.1. Methodological tendencies and unaddressed gaps

The dominance of quantitative and mixed-method approaches in the reviewed studies highlights a prevalent emphasis on measuring learning outcomes, such as decision-making skills, engagement levels, or critical thinking gains in controlled environments. While these approaches provide valuable empirical insights, it may overlook nuanced learner experiences, especially in culturally or contextually diverse educational settings. Although some studies refer to the experimental and quasi-experimental design, the absence of randomized controlled trials and clearly defined variables hinders the possibility of conducting comprehensive *meta*-analyses. Previous studies have largely neglected the need for more robust research designs, leaving room for future studies to refine methodologies and standardization of data. Similarly, Faisal et al. [17] also indicate a lack of standardized assessment methods for evaluating the BSGs. Notably absent are qualitative studies that could delve into how students interpret and emotionally respond to these simulations. This gap suggests an opportunity for future research to employ ethnographic, case study, or phenomenological approaches to uncover rich, contextual insights—particularly in classrooms adopting innovative technologies.

Moreover, the integration of VR, AI, blockchain, web-based (open sources) and mobile platforms into BSGs represents a significant shift in business education. Nevertheless, as advanced technologies gradually evolve, it is necessary to immerse students within BSGs integrated with other technologies like AR, IoT, blockchain, and XR. Here, it is important to emphasize that although AR and IoT were part of the search strategy, they were underrepresented in the final reviewed studies. Their inclusion in this discussion is aimed at outlining emerging research directions rather than summarizing current empirical findings. These technological advancements can harness their collective potential in shaping the future of business education. Similarly, Bach et al. [6] in their review highlight the need for more technology-driven, including AI, AR, or e-learning BSGs to improve students’ outcomes.

Moreover, a noticeable gap exists in the literature regarding the customization of BSGs to address diverse learning preferences, such as visual, auditory, and kinesthetic styles. This gap is particularly striking given the increasing focus on designing adaptable tools that respond to individual learner needs. Moreover, the issue of accessibility for students with disabilities remains underexplored, highlighting the need for further investigation into creating inclusive learning experiences.

From the aspect of geographical distribution, there is an underrepresentation of studies on BSGs in non-Western contexts, with few addressing necessary cultural adaptations. Building upon this result, Hernández-Lara et al. [26] also emphasize the significance of cross-cultural differences in business simulations, particularly in managerial skill development. The lack of research on cross-cultural adaptability limits the generalizability of findings to a global audience. Therefore, future research should prioritize cross-cultural studies to assess the global applicability of BSGs.

6.2. Learning gains and competency development in enhanced BSGs

While emerging technologies have enhanced the educational value of business simulation games (BSGs), much of the existing research examines these tools in isolation. The combined use of AI, VR, blockchain, and web-based or mobile platforms within a single simulation is still underexplored. It remains unclear whether integrating multiple technologies creates a more immersive and effective learning experience or leads to cognitive overload and fragmented learning. Future research should investigate how these tools interact when used concurrently and how they can be integrated in a pedagogically coherent manner. It is worth reiterating that the emphasis on integrated multi-technology systems—such as the joint use of AI, VR, XR and/or blockchain—stems from the broader scope of our review rather than the prevalence of such designs in the reviewed literature.

A further research gap concerns the alignment between technological features and specific learning objectives. Although many studies demonstrate the potential of BSGs to engage learners [11,13,41,73], few provide evidence on how these simulations support the development of targeted competencies, such as ethical decision-making, strategic thinking, or teamwork. On the other hand, while mobile and web-based platforms increase access, few studies explore how cultural, linguistic, or socio-economic factors influence student engagement and learning outcomes. Future research should prioritize inclusive design and accessibility to ensure BSGs support diverse learners across different educational contexts.

Additionally, unlike previous reviews [17,20], our study suggest that most studies assess short-term outcomes, leaving the long-term retention and transferability of skills unexamined. Thereby, longitudinal studies are needed to evaluate whether competencies developed through BSGs translate into real-world business performance over time. Moreover, traditional assessment methods may not adequately capture the complex, process-oriented skills developed in technology-enhanced simulations. There is a clear need for alternative assessment approaches that reflect the interactive, adaptive, and experiential nature of these learning environments.

6.3. Barriers to effective integration of emerging technologies in BSGs

This review shows that while business simulation games (BSGs) integrated with emerging technologies have some potential in terms of decision-making and skill development, a number of significant pedagogical and technical limitations still exist. The main problem is the lack of compatibility between technological innovation and teaching effectiveness. Previous studies [25,62] also suggest to conduct the effectiveness of pedagogical methods used in business simulation games. Simulations enriched with technologies such as AI, Generative AI chatbots, VR, and blockchain often do not provide comprehensive feedback, do not sufficiently reflect real-world scenarios, and require intensive teacher support. In particular, AI-based simulations have a weak impact on achieving interdisciplinary and complex outcomes (e.g., sustainable development, ethical decision-making) [63]. This situation is often associated with cognitive overload and poor pedagogical design [38]. These challenges were most commonly identified in studies that integrated AI, GenAI chatbots/agents, VR and XR. The limited number of studies involving blockchain, and the absence of IoT applications, restrict generalization of these findings to all forms of emerging technologies. Nevertheless, their mention reflects future pedagogical concerns should these technologies become more widespread.

The review analysis reveals a number of important research gaps. First, there is a lack of empirical research examining the interactions of low-achieving students with BSGs. Second, there is a lack of research measuring the long-term impact of these games on real-life skills. Third, the cultural and contextual adaptation of BSGs, especially in non-Western settings, has not been sufficiently studied. Fourth, most research focuses only on student outcomes and does not consider the role of teachers, as well as their instructional needs.

6.4. Pedagogical preconditions and contextual validity

This review deepens our understanding of emerging technology-integrated business simulation games by emphasizing the importance of pedagogical and contextual factors alongside technological capabilities. While existing studies often highlight the engaging features of emerging technologies such as AI, VR, and blockchain, this review reveals that their effectiveness largely depends on how well they are integrated into instructional design and supported by educators. In our final sample, most studies engaging with design and cognitive factors involved AI and VR simulations. Nonetheless, we include references to other technologies, such as blockchain, XR, to indicate directions for future instructional design research, even though their empirical basis remains limited.

A key finding is the relevance of cultural and contextual adaptation. Unlike previous approaches that treated BSGs as universally transferable, the reviewed studies suggest that student engagement is closely tied to the authenticity and contextual alignment of the scenarios. Simulations that reflect local business dynamics and culturally relevant entrepreneurship scenarios are more likely to foster strategic thinking and intrinsic motivation. In contrast, simulations lacking realism or cultural relevance may lead to surface-level engagement or trial-and-error learning strategies. Regarding Vygotsky's [75] Constructivist Learning theory, it is crucial that students in simulation games collaboratively solve real-world problems, developing critical thinking and business competencies through socially mediated and context-rich experiences. Our review findings contrasts with more generic approaches found in prior research, which may overlook the influence of regional norms and teaching practices [18,72].

The review draws attention to the underexplored area of teacher preparation and support. Although previous research tends to prioritize technological affordances with teachers' support in traditional learning settings [62], the review shows the successful use of BSGs depends heavily on instructor facilitation and instructional strategies. This finding also aligns with the Vygotsky's [75] Constructivist Learning theory, which indicates that teacher support (scaffolding) facilitates student learning by guiding them through tasks they are not yet able to complete independently. As a result, we point out that future research could explore practical models for teacher training and curriculum integration tailored to specific technologies.

Moreover, the study highlights a shift from external incentives such as rewards or points to an emphasis on intrinsic motivation, self-regulation, autonomy and self-management. This shift is explained by the notion that extrinsic incentives are limited to generating short-term interest and undermine intrinsic motivation and autonomy, thereby hindering long-term learning. From this aspect, Hendijani et al. [27] also indicate that relying on extrinsic incentives can reduce learners' intrinsic interest and lead to superficial learning. Conversely, promoting autonomy and intrinsic motivation promotes active and sustained engagement. This approach is consistent with self-determination theory [14] which emphasizes learner agency, real-life applications and the development of lifelong skills driven by intrinsic motivation.

6.5. Technology-specific design considerations

We observed that current research on the types of AI technologies, in particular, GenAI (agents/chatbots) integrated into BSGs is still relatively rare. This calls for adding the different types of AI, such as GenAI to enrich simulations. We suggest to explore new AI applications – for example, systematic investigations of GenAI agents as adaptive virtual instructors or opponents – and measure their impacts on learning outcomes. Research should focus on making AI feedback more comprehensive (covering ethics/sustainability) and critically examine its effects on students' independent cognitive abilities and confidence.

Immersive (e.g., VR-based, XR-based, etc.) BSGs have attracted interest for their realism and engagement, but important questions remain. Some studies show VR boosts motivation and risk-awareness, yet others report potential downsides such as cognitive overload or reduced performance under high immersion. Therefore, there is a need to optimize interface design (e.g. more intuitive UI/UX and sensory labeling) and to understand which learners benefit most.

Mobile and web-based BSGs are widely accessible, but their pedagogical potential and limitations are not fully understood. Web-based games are typically easy to deploy and can incorporate gamified elements, yet many studies show their lack of realism and declining engagement over time. Future work should explore how to increase authenticity in web simulations (e.g. more dynamic scenarios, richer data) and study how to sustain motivation (for example, by integrating adaptive narratives or social features). Likewise, research is needed on effective instructional scaffolding for mobile games. From this aspect, it can be interesting to analyse what guided teaching methods or team-based activities enhance learning when using mobile simulations.

Emerging areas like blockchain- or "Web3"-enabled simulations (for example, tokenized trading games or transparent supply-chain models) are largely unexplored. The few existing prototypes (e.g. token-based supply-chain games) suggest promise in teaching decentralized trust and digital asset management, but practical implementation is complex. Key gaps include designing intuitive blockchain mechanics for students and empirically testing whether blockchain gameplay truly builds understanding of decentralized systems. Researchers should investigate pedagogical strategies for blockchain – for example, scaffolded tutorials on blockchain concepts or real-world case studies – and measure learning effects.

All in all, this review suggests a shift in research and practice—from focusing primarily on technological innovation to prioritizing its meaningful integration into pedagogy. While this study included technologies such as blockchain and IoT in its initial scope, the limited number of studies on these topics necessitates caution in drawing generalizations. Future research should therefore pursue both broader empirical inclusion and deeper pedagogical theorization to support emerging technologies in BSGs. Future research should move beyond assessing the novelty of technologies to explore how they can be pedagogically grounded, culturally contextualized, and equitably accessed across diverse learner groups.

6.6. Summary of the review findings

Each category below (Table 6) summarizes how emerging technologies contribute to the BSGs while also highlighting their challenges. For example, AI-powered and GenAI-based BSGs can deliver tailor-made feedback and complex scenarios, but researchers must address the narrow focus of current AI systems and study long-term learning effects. Similarly, VR-based simulations create realistic practice fields but require better interface design and careful integration into curricula to avoid cognitive overload. Web and mobile platforms excel at accessibility and engagement, yet their pedagogical design needs refinement (e.g., increasing task complexity or aligning with learning objectives). Blockchain-based games hold promise for teaching trust and digital economy concepts, but their novelty means we need more experimental

Table 6
Synthesis of key findings.

Technology types	Key educational benefits	Key limitations	Learning outcomes and impacts
AI / GenAI (e.g. Chatbots)	Personalized real-time feedback; adaptive tasks; student-centered learning; enhanced decision-making in complex scenarios	Limited feedback scope (narrow domains); risk of over-reliance hindering independent thinking; high technical complexity and instructor support needed	Stronger analytical/strategic thinking; higher entrepreneurial mindset and confidence; improved problem-solving and adaptive skills
Virtual reality	Immersive, realistic business environments; safe experimentation; higher engagement; enhanced risk assessment	Technical/equipment barriers; potential cognitive overload; perception of low realism or immersion gaps; UX/UI design challenges	Improved complex decision-making; increased critical thinking and adaptability; better understanding of spatial/contextual business skills
Web-based Simulations	Easy accessibility; gamification engages learners; supports collaboration; widespread adoption	Perceived lack of realism leading to random play; limited external validity; engagement may wane over time	Knowledge acquisition, enhanced strategic and critical thinking; improved market and financial literacy; increased motivation and self-directed learning
Mobile Simulations	Portability and anytime access; interactive features (e.g. avatars, gamified UI); builds entrepreneurial attitudes	Often shallow content depth; requires structured instructional support; gains in attitude may not translate to changed intentions; device constraints	Higher self-efficacy and interest in business topics; better soft skills (confidence, innovation); increased motivation, especially for informal learning
Blockchain-enabled	Introduces decentralized asset ownership and trust; hands-on with token-based economies; transparent digital credential tracking	Steep learning curve and abstraction; complexity may frustrate novices; requires scaffolding to be pedagogically meaningful	Enhanced understanding of digital finance and trust in transactions; strategic thinking about supply chains and digital resource management; engagement with emerging digital competencies

evidence on how such games affect learning. In all cases, future research should combine these technology insights with robust study designs and broader outcome measures. By addressing these gaps – through comparative studies, longitudinal tracking, inclusion of diverse learners, and richer analytics – the field can build on current findings and ensure that simulation-based learning reaches its full potential in business education.

7. Conclusion

This study synthesized findings from 46 empirical studies to analyze the role of emerging technologies (VR, AI, Generative AI chatbots, blockchain technologies, mobile-based and web-based technologies) into business simulation games (BSGs). This review has several contributions for the research community.

This review underscores the transformative potential of emerging technologies in business simulation games (BSGs) while highlighting critical gaps that must be addressed to maximize their educational value. Business simulation games supported by emerging technologies like AI, generative AI, VR, and blockchain, along with web-based or mobile simulation games offer valuable opportunities for developing students' business skills. However, their success depends not just on the technology itself, but on how well it is used in teaching. This review shows that BSGs are most effective when they are designed to match learners' needs, reflect real-world and cultural contexts, and include support from teachers. Future research should focus on making these tools more inclusive, practical, and relevant to long-term learning. By doing so, BSGs can better prepare students for the challenges of modern business environments.

Although the review analysed different dimensions related to emerging technology-integrated BSGs, some methodological limitations still need to be acknowledged. The analyses might have not covered all of the relevant studies apart from the reviewed studies, even if the study content aligned with the scope of our research. This case can occur since they might not explicitly have mentioned research focus in their titles, abstract, or keywords. Moreover, we mostly relied on studies and conference proceedings written in English. As a result, the review excluded findings of theses, dissertations, or book chapters, along with articles written in languages other than English. Future review studies should consider to broaden the review scope. The review is based on published works from selected indexed databases. Future research could broaden the review to incorporate additional indexed databases. On the other hand, we acknowledge that it is inevitable to completely avoid biases in article selection procedure, similar to previous reviews (e.g., [46]. It is an irrefutable fact that this case occurs when studies with positive or significant results are more likely to be published, leading to a distorted research landscape and an overestimation of the effectiveness of certain interventions or approaches. Thus, we recommend for the future review studies using broader search criteria, applying advanced search technologies, and promoting international collaboration in future studies. This process can ensure more robust methodological approach for the new review studies.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.entcom.2025.101006.

Data availability

The datasets analyzed in this review are included in the manuscript. Additional details are available from the corresponding author upon request.

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Articles marked with an asterisk (*) are included in this review.

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